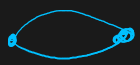
Graph Theory

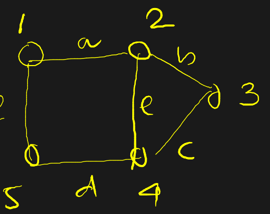
Set of vertex/nodes

$G(V, E)$ → edges

- ✓ Adjacent vertex → Common edge → {1,2} ✓ {1,3} ✗
- ✓ Adjacent edge → Comm vertex – {a,b} ✓ {a,d} ✗
- ✓ Self loop (SL) –

- ✓ Multi edges Parallel (||)





$G(V,E)$

$V = \{1,2,3,4,5\}$

$E = \{a,b,c,d,e\}$

- ✓ Pseudograph → Graph contains self loop ✓ + Multi edges ✓.
- ✓ Multigraph — SL → ✗ , || → ✓
- ✓ Simple Graph – SL → ✗ , || → ✗.

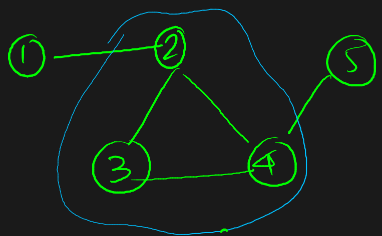
$\boxed{\text{Multigraph} \subseteq \text{Pseudograph}}$

Formula

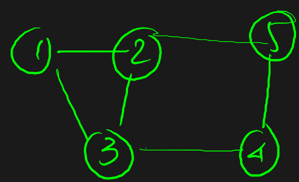
$$\boxed{\sum \deg(V) = 2E}$$ ✓ → b/c every edge touches 2 vertices.

Path → seq of vertices connected by edges. A → B → C → D.

Cycle → Cycle is a path that starts and ends at same vertex. $\boxed{A}$ → B → C → $\boxed{A}$

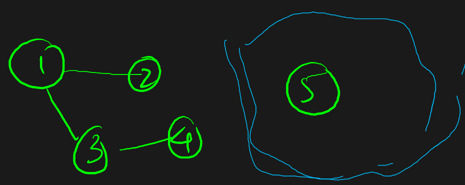


Connected graph → Every vertex can reach every other vertex. →



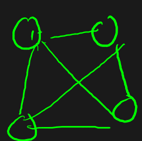
Complete

Disconnected graph → Some nodes are isolated



Disconn

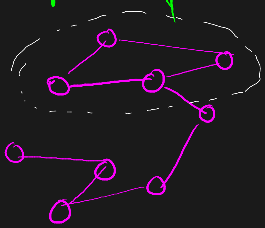
Complete graph → Every vertex is conn. to every other vertices.



Connected

$$\boxed{E = \frac{n(n-1)}{2}}$$ n = # vertices

Subgraph — A graph formed from part of another graph



Tree - Special graph that is — connected
— no cycles.

$\boxed{E = n-1}$

n → # vertices.

Final Quick Revision

Topic	Main Idea
Graph	Nodes + connections
Vertex	Node
Edge	Connection
Directed graph	Has arrow direction
Undirected graph	No direction
Weighted graph	Edge has value
Degree	Number of connected edges
Path	Route between vertices
Cycle	Returns to start
Connected graph	Every node reachable
Complete graph	Everyone connected to everyone
Subgraph	Part of graph
Tree	Connected + no cycles

MOST IMPORTANT THINGS TO REMEMBER

1. Degree formula:

$\sum \deg(v) = 2E$

2. Tree formula:

$E = n - 1$

3. Complete graph formula:

$E = \frac{n(n-1)}{2}$

Spanning Tree (ST)

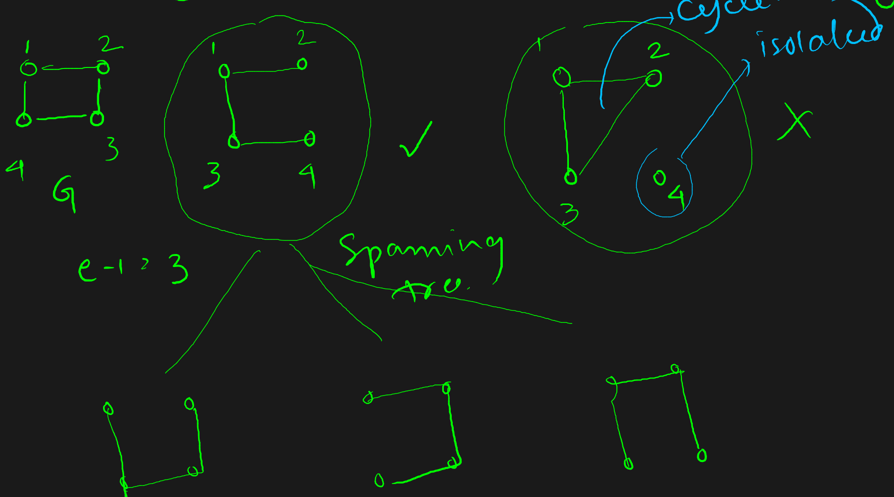
$S \subset G(V,E)$

↓

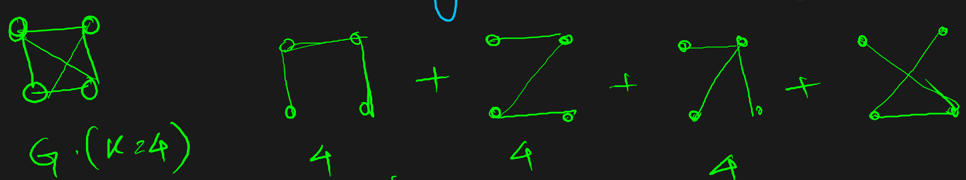
A spanning tree if ⟶

① S should contain all vertices of G.

② S should contain $|V|-1$ edges.



Q. Given $K=4$ complete graph. # spanning trees = ??



= 16 ✓ Total ST

4 4 4

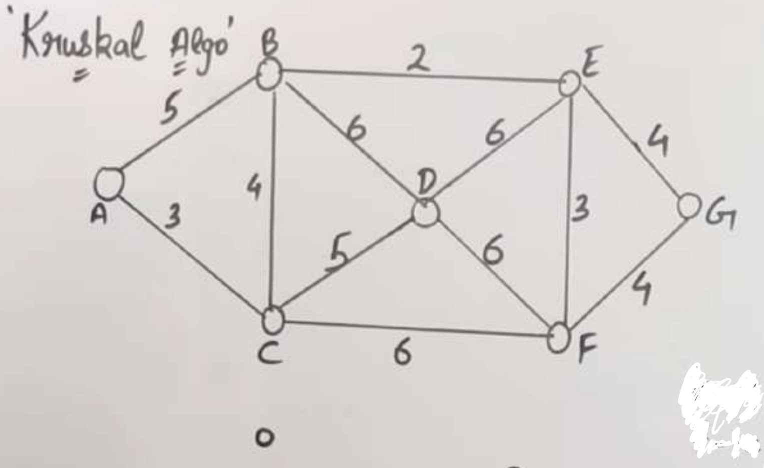
Rotate in 4 directⁿs.

Formula : For complete graph :

$\boxed{\# ST = n^{n-2}}$ n = # vertex.

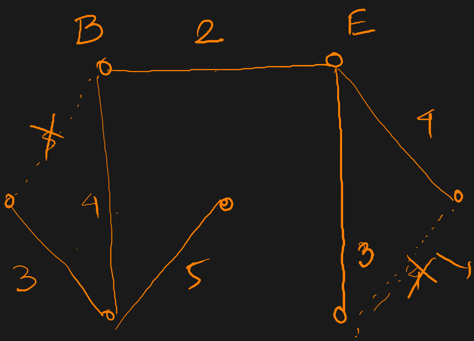
$4^{4-2} = 4^2 = 16$ ✓

Kruskal Algo



1) Construct Min Heap with 'e' edges
2) Take one by one edge and add in spanning tree. (Cycle should not be created)

Best Case (n-1) edges
Worst Case 'e' edges



$(n-1) = 7-1 = 6$ edges.

→ Place the edges with increasing weights.

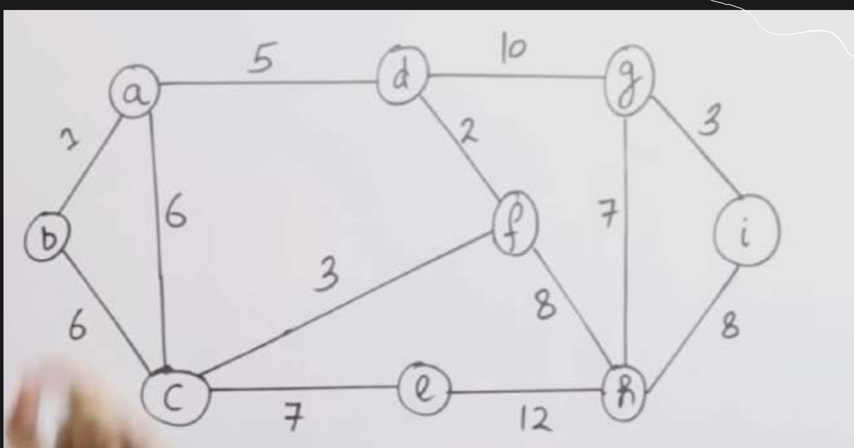
Cycle (do not consider)

↳ 6 edges.

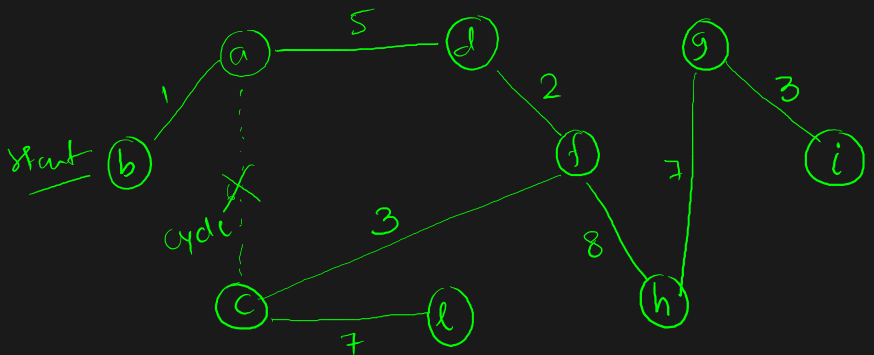
$3+4+5+2+4+3+4 = 21$ → Minimum Cost ST.

Prim's Algo →

Minimum Cost ST.



① Start from any vertex
② Choose min cost among all edges.



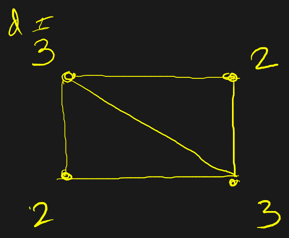
→ # edges = $n-1 = 9-1 = 8$.

→ Min ST (MST)

Total cost = $1+5+2+3+7+8+7+3 = 36$

Handshaking Lemma

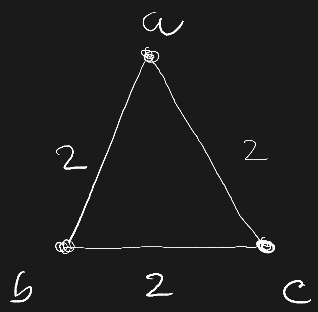
Max degree of any vertex in a simple graph with n vertices is $(n-1)$



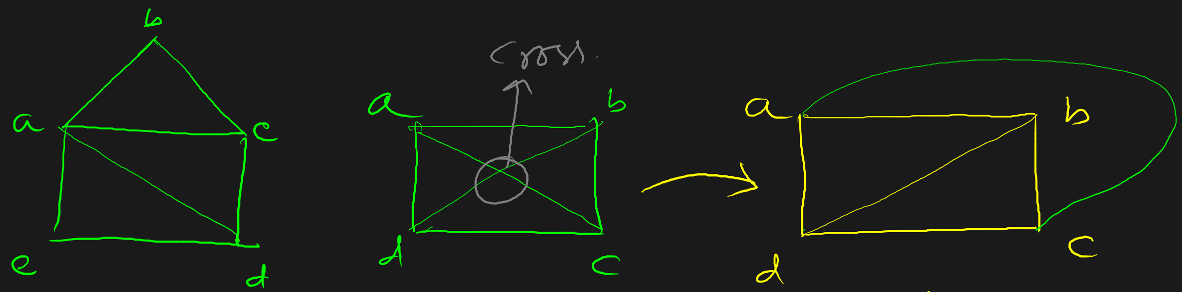
$$\boxed{\sum_{i}^{n} d(v_i) = 2e}$$

$3+3+2+2 = 10 = 5 \times 2$ → Proved

5 → e.

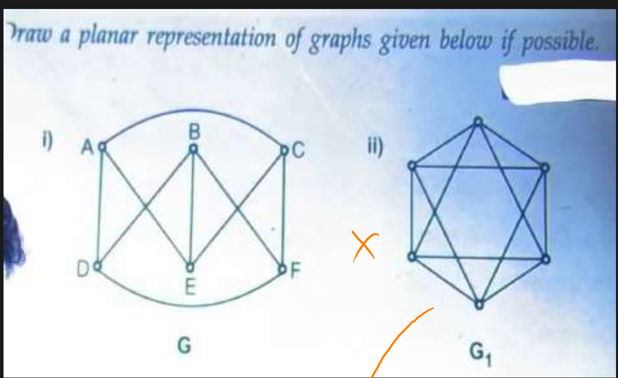


Planar Graph: A graph is said to be planar if it can be drawn on a plane in such a way that no edges cross one another.



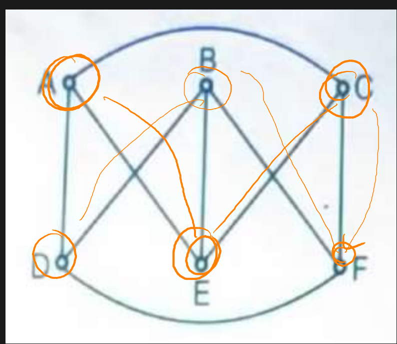
planar ✓ × ✓

Q.



Draw a planar representation of graphs given below if possible.

(i)



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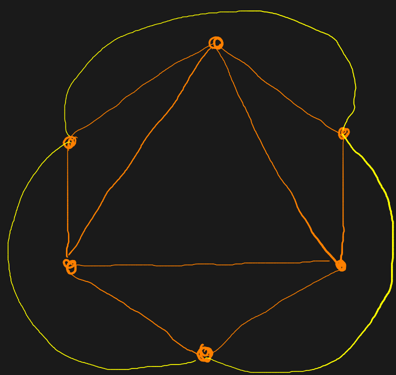
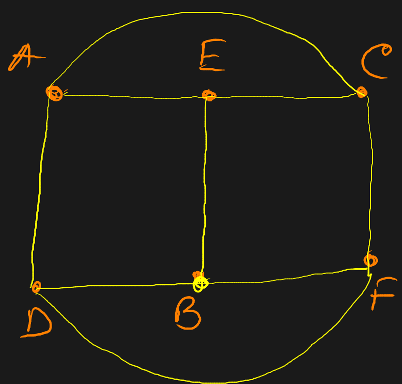


✓

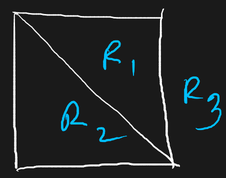


✓

Euler's formula

$$\boxed{V - E + R = 2}$$

Region



Q:

Show that in a connected planar graph with 6 vertices 12 edges each of region is bounded by 3 edges.

$V = 6$
$E = 12$
$R = ?$

Euler's Formula $V - E + R = 2$

$6 - 12 + R = 2 \quad \therefore \boxed{R = 8}$

So, total we have 8 edges. & each region bounded by 3 edges.

Now,

→ 1 edge form 2 regions.

✓ \ ✓

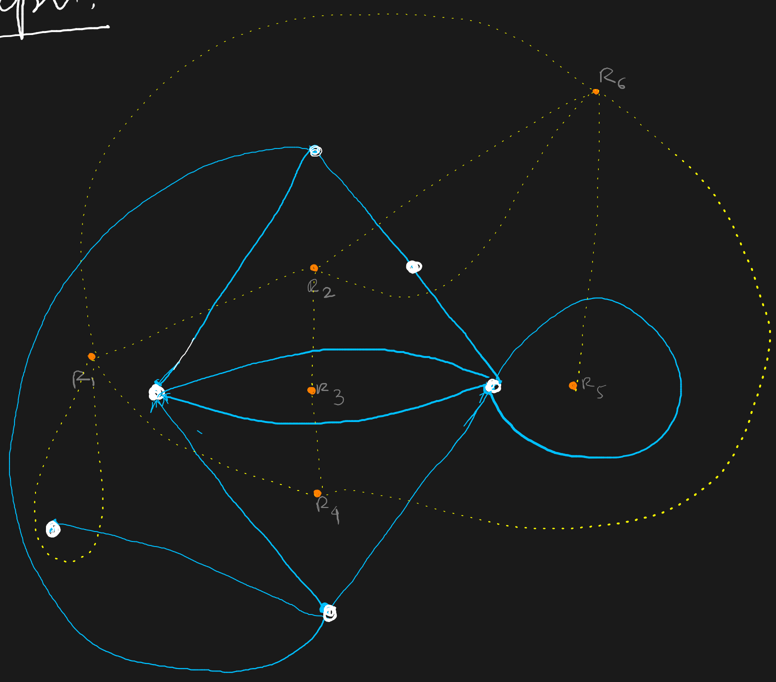
$\therefore$ 12 edges form $12 \times 2 = 24$ regions.

here $R = 8$

Proved

$\therefore$ For making 1 region required edges $= \frac{24}{8} = \boxed{3}$

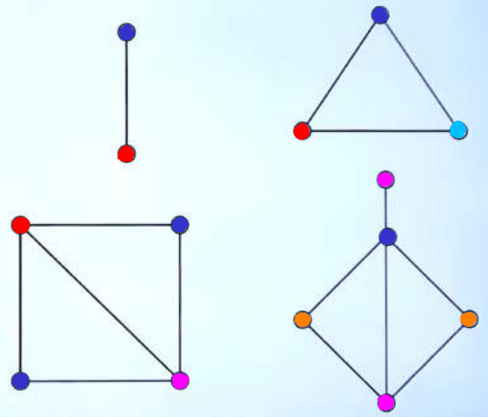
Dual of A Graph.



GRAPH COLORING

Painting all the vertices of a graph with colors such that no two adjacent vertices have the same color is called coloring of graph.

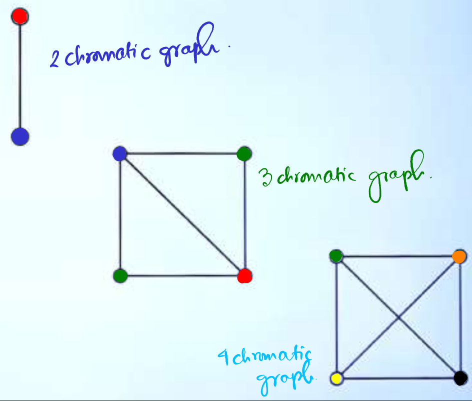
Example :



CHROMATIC NUMBER

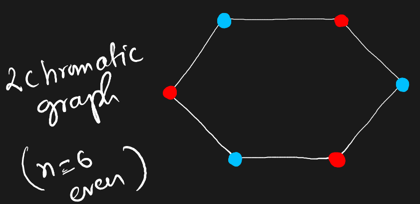
The least number of colors required for coloring of a graph G is called its chromatic number.

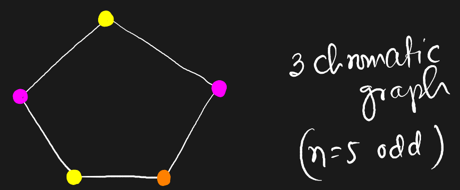
Example :



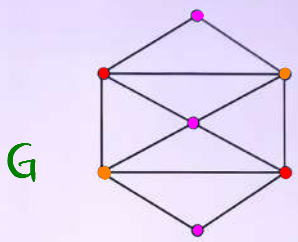
1. chromatic no. of graph G $\to \chi(G)$
2. $\chi(G) = k \to k$ chromatic.
3. χ of null graph $= 1$
4. $\chi(G)$ where $G\ (K=n)$ then $\chi(G) = n$

5. If a graph is circuit with n vertices then -
n even → 2 chromatic
n odd → 3 chromatic.



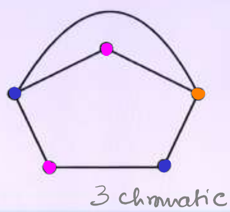


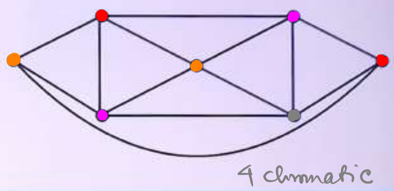
2 Find the chromatic number of graph.



$\chi(G) = 3$

3 What are chromatic numbers of the graph G & H as shown in figure.



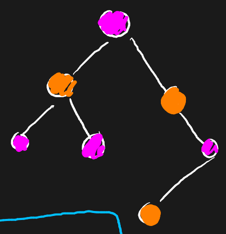


4 Find the chromatic number of graph

3 chromatic

Chromatic no. → graph → Even cycle → $\chi(C_n) = 2$
odd " → $\chi(C_n) = 3$

Trees : $\chi(T_n) = 2$ always.



Clique → Complete Subgraph of a graph. $\boxed{\omega(G) \leq \chi(G)}$

Clique number.

Chromatic no. can never be smaller than largest clique.

Meaning
If a graph contains Δ then the graph needs ≥ 3 colors.

$\boxed{\Delta(G)}$ → Maximum degree
→ largest degree among all vertices.

Upper bound theorem → $\boxed{\chi(G) \leq \Delta(G) + 1}$

if maximum degree = 5 then graph can be colored using atmost 6 colors.

Brooks' Theorem

Statement : for connected graph $\boxed{\chi(G) \leq \Delta(G)}$ except:
- complete graphs
- odd cycles.

usually we do not need $\Delta + 1$
only Δ colors are enough.

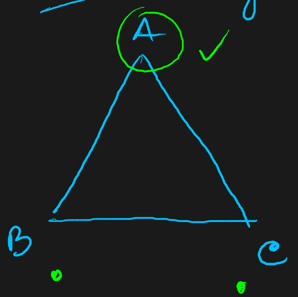
Eg: $\Delta(G) = 5$ then often $\chi(G) \leq 5$ → 6 not required.

Exception : complete graph : needed $n = \Delta + 1$ colors.
odd cycle : needed 3 but $\Delta = 2$ $\therefore (\Delta + 1)$ colors.

Chromatic Polynomial : no. of valid colorings using x colors. $\boxed{P(G, x)}$

✓ Chromatic no. → # of colors needed
✓ Chromatic polynomial → # of ways to color.

Eg: Triangle.



suppose for 1st vertex # of choices = x.
$\therefore$ for 2nd vertex = $(x-1)$, 3rd = $(x-2)$

$\therefore P(K_3, x) = x(x-1)(x-2) = \boxed{x^3 - 3x^2 + 2x}$

Chromatic Polynomial of TREEs

For tree with n vertices → T_n → $\boxed{P(T_n, x) = x(x-1)^{n-1}}$

because → For the 1st vertex suppose $= x$.
Next every vertex cannot match parent so, for all $(x-1)$

$$\therefore \underbrace{x}_{1} \cdot \underbrace{(x-1)\cdot(x-1) \cdots\cdots (x-1)}_{(n-1)} = x \cdot (x-1)^{n-1}$$

For a tree with 5 vertices → 5×4^4 → rest all 4 choices.
5 → For 1st vertex 5 choices

Properties of Chromatic Polynomial

1. No constant term → Because 0 colors impossible.
2. For connected graph $P(G,1) = 0$ → b/c 1 color usually insufficient.
3. Chromatic no. is smallest +ve integer where $\boxed{P(G,k) > 0}$.

→ Eg: if $P(G,1) = 0$
$P(G,2) = 0$
$P(G,3) > 0$
then, $\chi(G) = 3$

Decomposition Theorem (Deletion–Contraction)

→ to simplify difficult graph coloring problems.

$$\boxed{P(G,x) = P(\boxed{G-e}, x) - P(\boxed{G/e}, x)}$$

remove edge 'e'

contract edge, merge its 2 vertices

MOST IMPORTANT FORMULAS SUMMARY

Complete Graph
$\chi(K_n) = n$

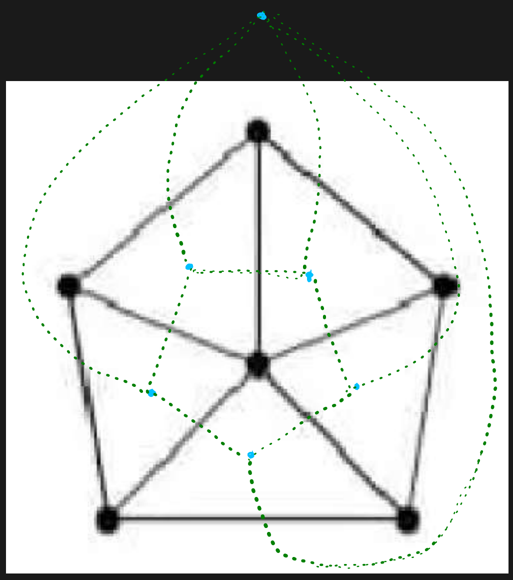
Tree
$\chi(T_n) = 2$

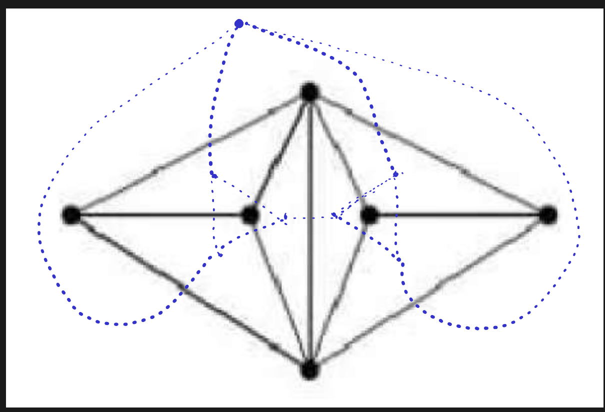
Even/Odd Cycle
$\chi(C_n) = \begin{cases} 2, & n \text{ even} \\ 3, & n \text{ odd} \end{cases}$

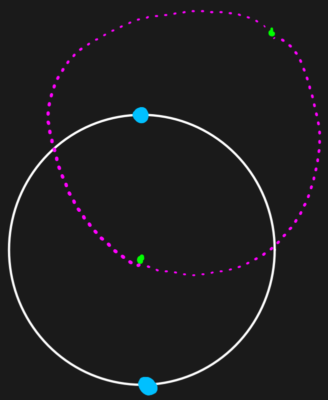
Tree Polynomial
$P(T_n, x) = x(x-1)^{n-1}$

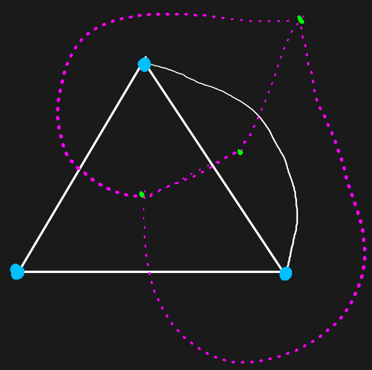
Triangle Polynomial
$P(K_3, x) = x(x-1)(x-2)$

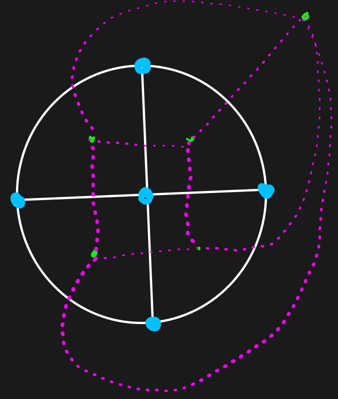
Questions on Dual of a Graph.





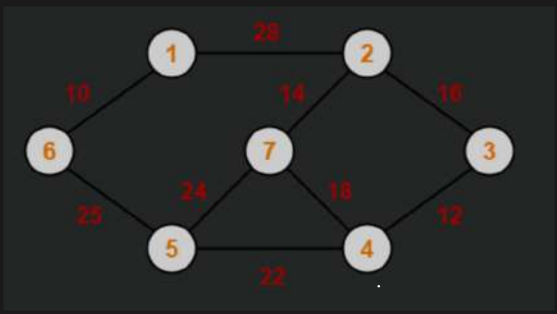




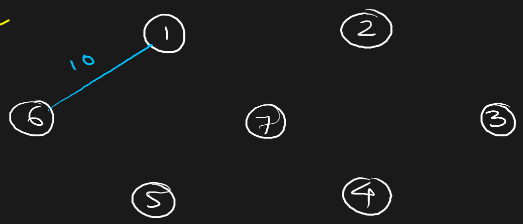


Questions on Prims' & Kruskal algo.

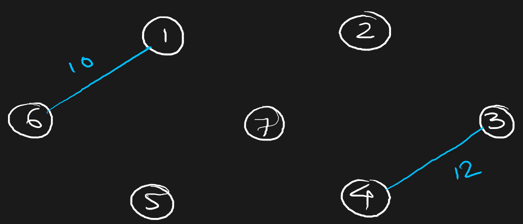
Kruskal



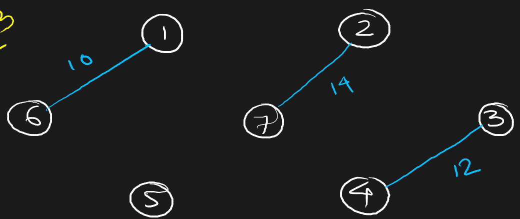
Step1



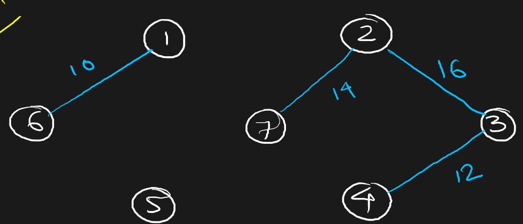
Step2



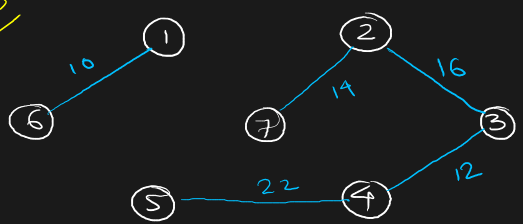
Step3



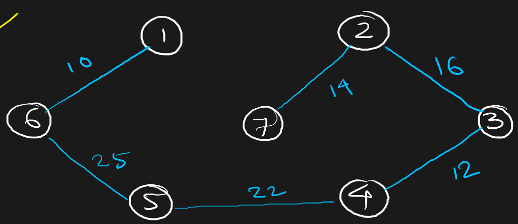
Step4



Step5

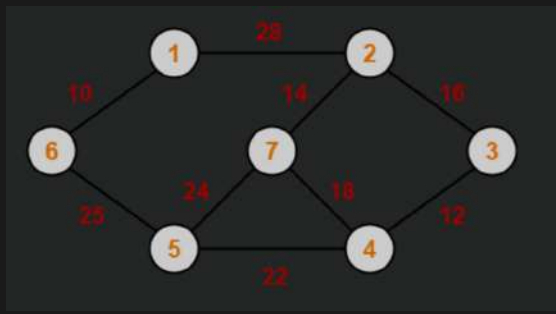


Step6

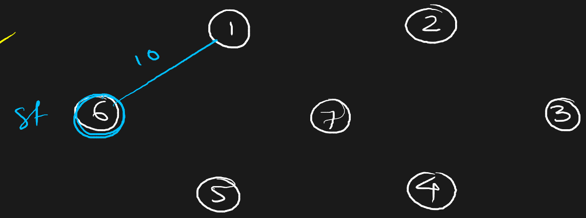


Cost = 10+25+22+12+16+14 = 99

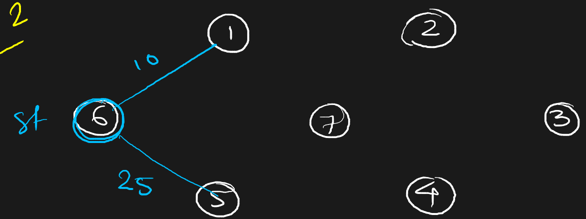
Prims.



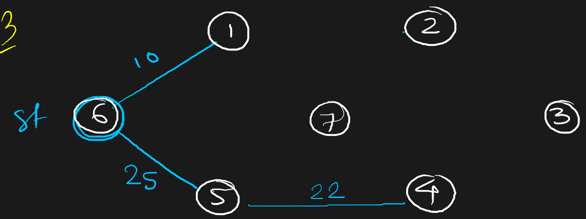
Step1



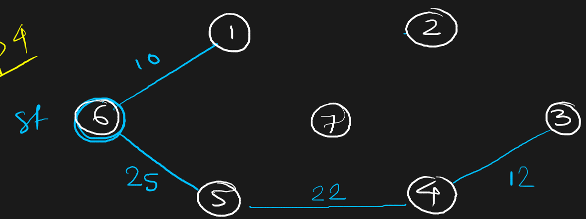
Step 2



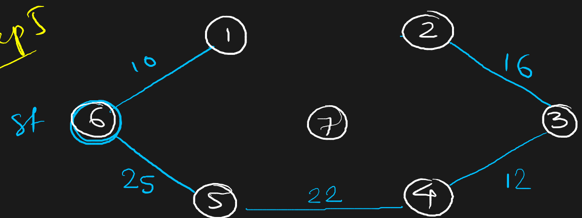
Step3



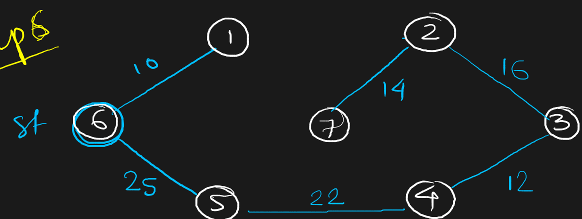
Step4



Step5



Step6



Cost = 99.